

Vinh Ly

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Summary

Software Engineer with **18 years** building consumer products and internal platforms at scale. Led end-to-end ownership of systems serving millions of daily active users including YouTube Shopping, Google Meet, and a live marketplace platform with **20M+ concurrent users**. Managed teams of up to **15 engineers** across product and infrastructure work. Partners with product and design to define technical roadmap and delivery. Works on product engineering, developer tooling, and internal platforms with emphasis on cloud-native architectures. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages: Ruby on Rails, Go, Python, Java, C++, TypeScript, JavaScript, HTML, CSS

Infrastructure: Distributed Systems, Kubernetes, GCP, AWS

Data: PostgreSQL, Redis, Kafka

Platform: CI/CD, Internal Developer Tooling, API Design, On-call

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Lead API design for new creator monetization endpoints, partnering with product and mobile teams to define contracts and rollout strategy.
- Own video infrastructure serving **20M+ concurrent users**, scaling real-time streaming pipelines and managing on-call rotations for zero-downtime deployments.
- Drive cross-team alignment on video delivery roadmap, coordinating with Android, iOS, and web frontend teams on feature parity.
- Reduce infrastructure costs by 15% through migration to regional endpoints and optimized resource allocation across **GCP**.
- Mentor senior engineers on distributed systems patterns and conduct technical interviews for pipeline scaling.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Built seller onboarding and live auction infrastructure handling **millions of daily transactions** on **Ruby on Rails** and **Go** microservices.
- Led migration of core payment paths to **PostgreSQL** and **Redis** caching layer, cutting p99 latency from 800ms to 120ms.
- Defined technical strategy for real-time bidding system, presenting to leadership and aligning team on 6-month roadmap.
- Established on-call runbooks and SLO tracking for marketplace platform serving **millions of concurrent users**.
- Shipped two major product launches including mobile push notification infrastructure with delivery rates above 99.5%.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Managed team of **15 engineers** across YouTube Shopping and payment infrastructure, conducting performance reviews and career development for 6 direct reports.
- Led product engineering for collaboration tools reaching **hundreds of millions of users**, driving roadmap prioritization and cross-functional alignment.
- Delivered migration of legacy data pipelines to **Kubernetes**-based workflows, reducing processing latency by 40% and cutting infrastructure spend.
- Established internal platform standards including API design guidelines and code review practices adopted across 3 engineering teams.

- Negotiated technical contracts with external partners, translating business requirements into system designs with clear SLAs.

Software Engineer | Google | 02/2011 - 02/2015

- Built internal tooling for **C++** and **Java** codebases used by 500+ engineers, reducing build times by 25% through parallelization.
- Optimized **distributed systems** performance under peak load, profiling bottlenecks and implementing caching strategies reducing DB load by 30%.
- Designed data storage solutions for large-scale information retrieval, managing **on-call** escalations and incident response for production systems.
- Led migration of internal web applications from legacy stack to **Python**-based services with **PostgreSQL** backends.
- Automated testing frameworks for core platform services, achieving 95% test coverage across critical code paths.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Developed internal web applications using **C#** and **.NET** framework, delivering automation tools used by 200+ internal teams.
- Built automated testing infrastructure for product releases, reducing QA cycle time from 2 weeks to 3 days.
- Led integration of third-party services into internal platforms, managing vendor relationships and technical requirements.
- Contributed to **C++** backend services handling high-throughput data processing with strict latency requirements.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Collaborated with full-time engineers on bug triage and resolution, documenting defects and verifying fixes in **C#** codebase.
- Developed testing automation scripts for product validation, improving test coverage by 15% across key feature areas.

Education

B.S. Honors Degree in Computer Science

Virginia Tech | 2004 - 2007